

Programme	B. Sc. Mathematics Honours			
Course Code	MAT7CJ403			
Course Title	ABSTRACT ALGEBRA II			
Type of Course	Major			
Semester	VII			
Academic	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	3	2	75
Pre-requisites	1. Mathematical Logic and necessary exposure to set theory. 2. First Course on Group Theory			
Course Summary	The subject of group theory is taken upon from where it was left off in previous introductory courses. The basic constructions in group theory – those of direct products and quotient groups are introduced. The Fundamental Theorem of Finitely Generated Abelian Groups is introduced (without proof) and the consequences explored in order to compare the challenges in the theory of Abelian vs non-Abelian groups. After an introductory delving into normal and subnormal series of groups, group actions are introduced and Sylow Theory discussed in the context of classifying non-Abelian groups. The course concludes with a basic discussion on polynomial rings and their factorisation, paving the way for the theory of extension fields in later, more advanced courses.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply the concept of direct products of groups and factor groups to construct new groups from existing ones.	Ap	P	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
CO2	Analyse and evaluate the isomorphism theorems, series of groups, and Sylow theorems to understand the structural properties and classifications of groups.	E	C	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
CO3	Synthesize the concepts of rings of polynomials, factorization of polynomials, and ideal structures within rings and fields, with a focus on homomorphisms and factor rings.	E	P	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook		A First Course in Abstract Algebra, J. B. Fraleigh, 7th Edition, Pearson Education Limited, 2014.		
Module	Unit	Content	Hrs (45+30)	External Marks (70)
I	Basic Constructions – New Groups From Old		11	Min.15
	1	Section 11 – Direct Products of Groups (11.1 to 11.11)		
	2	Section 11 – Finitely Generated Abelian Groups (11.12 to 11.17)		
	4	Section 14 – Factor Groups		
	5	Section 15 – Factor Group Computations (15.1 to 15.13)		
	6	Section 15 – Simple Groups, The Centre and Commutator Subgroups (15.14 to 15.21).		
II	Advanced Group Theory (Pre-requisites: Sections 16 and 17 of Practicum)		14	Min.20
	7	Section 34 – Isomorphism Theorems		
	8	Section 35 – Series of Groups - 35.1 to 35.19 (Proofs of Zassenhaus Lemma and Schreier Theorem are optional)		
	9	Section 36 – Sylow Theorems (36.1 to 36.4)		
	10	Section 36 – Sylow Theorems (36.5 to 36.13).		
	11	Section 37 – Applications of the Sylow Theory (37.1 to 37.6)		
	12	Section 37 – Further Applications (37.7 to 37.15)		
III	Rings and Fields		11	Min.15
	13	Section 22 – Rings of Polynomials – (22.1 to 22.3) (proof of Theorem 22.2 is optional)		
	14	Section 22 – The Evaluation Homomorphisms (22.4 to 22.11)		
	15	Section 23 – Factorisation of Polynomials over a Field (23.1 to 23.6)		

	16	Section 23 – Irreducible Polynomials (23.7 to 23.21)		
	17	Section 24 – Non-commutative Examples. (24.1 to 24.3)		
	18	Section 24 – Non-commutative Examples (24.4 to 24.10)		
IV	More Ring Theory		8	Min.10
	19	Section 26 – Homomorphism and Factor Rings (26.1 to 26.6).		
	20	Section 26 – Factor Rings (26.7 to 26.19)		
	21	Section 27 – Prime and Maximal Ideals (27.1 to 27.20).		
	22	Section 27 – Ideal Structure in F[x] (27.21 to 27.27)		
V	Practicum: The goal is for the students to learn the following selected topics in 5 practicum sessions of six hours each via self-study and group activities. The lecturer may assist by running group discussions, supervising class seminars and referring library books for self-study and note preparation.		30	-
1	Section 12 – Plane isometries			
2	Section 16 – Group Action on a Set			
3	Section 17 – Application of G-sets to Counting			
4	Section 21 – The Field of Quotients of an Integral Domain			
	Section 35 - Series of Groups - Ascending central series - 35.20 to 35.21			
5	Section 39 – Free Groups			
References				
1. Abstract Algebra, Dummitt and Foote, Wiley India, 2011. 2. Contemporary Abstract Algebra, Joseph A. Gallian, CRC Press, 1986. 3. Topics in Algebra, I. N. Herstein, John Wiley and Sons, 2006. 4. Algebra, T. W. Hungerford, Springer-Verlag, 1987. 5. Algebra, Micheal Artin, Birkhauser, 2011 6. Algebra, Serge Lang, Springer, 2002. 7. Advanced Higher Algebra, J G Chakravorthy and P R Gosh, Kolkata U N Dhur, 2014 (ISBN: 9789380673059)				
Suggested Programming Exercises for Practicum:				
1. Form congruence groups Z_3, Z_2 . Verify that $Z_3 \times Z_2 \cong Z_6$. Form its				

- cosets (Section 9.11, Ref (3)).
2. Find the centre of the dihedral group. (Section 9.12, Ref (3))
 3. For an element from the dihedral group, find its stabilizer. (Section 9.12, Ref (3))
 4. Find the conjugacy classes of an element from the dihedral group. (Section 9.12, Ref (3))
 5. Take a subgroup (say H) of S_3 . List the conjugacy classes using the command conjugacy classes subgroups (). Can you find out all the subgroups using these conjugacy classes? (Ref (1) or Section 9.12, Ref (3))
 6. Find Sylow-2-subgroups and Sylow-3-subgroups of D_{18} (Section 9.13, Ref (3))

References

1. Robert A. Beezer; Group Theory and SAGE: A Primer, <http://people.reed.edu/~davidp/332/sage-group-theory.pdf>
2. Group Theory and Sage - SageMath tutorial https://doc.sagemath.org/html/en/thematic_tutorials/group_theory.html
3. Ajit Kumar, Vikas Bist; Group Theory An Expedition with SageMath, Narosa Publishing House.
4. Thomas W. Judson, Robert A. Beezer; Abstract Algebra Theory and Applications with Sage Exercises for Abstract Algebra, <http://abstract.ups.edu/download/aata-20130816.pdf>

***Optional topics are exempted for end semester examination.**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	0	3	0	2	0	1
CO 2	2	3	1	2	3	0	3	0	3	0	2
CO 3	2	1	3	3	3	0	3	0	3	0	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/ Report	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓